# V01 ACTIVE BUILDINGS AND COMMUNITIES | P

**Intent :** Facilitate all types of movement, including physical activity and exercise and reduce sedentary behavior through the intentional design of built spaces.

**Summary :** This WELL feature requires projects to select from a series of design-based optimizations.

**Issue :** Physical inactivity is linked to premature mortality and chronic diseases, including type II diabetes, cardiovascular disease, depression, stroke, dementia and some forms of cancer.[1–3] Despite widely disseminated physical activity guidelines (Appendix V1), global estimates from 2016 show that nearly a quarter (23%) of the adult population are physically inactive.[3] In addition, our homes, schools, workplaces, communities, jobs and transportation systems have been physically designed to demand less movement and require more sedentary activities.[3,4]

**Solutions :** We have come to understand that our environments play a significant role in physical activity behaviors. [2,5–10] Active design considers how different components of a building, such as staircases can encourage movement.[11] At a community scale, active design considers the ways in which communities can encourage populations to be active through public infrastructure, such as cycle lanes and green space. These factors, beyond physical activity promotion, also positively impact environmental, social and economic outcomes.[11,12] The impact of changing the global physical activity narrative is substantial. Worldwide, if physical inactivity were reduced by just 10%, more than half a million deaths could be averted, while over one million deaths could be averted, if physical inactivity were reduced by 25%.[13]

## PART 1 DESIGN ACTIVE BUILDINGS AND COMMUNITIES

*For All Spaces:*

The project achieves at least one point in one of the following features:

a. Feature V03: Circulation Network.

b. Feature V04: Facilities for Active Occupants.

c. Feature V05: Site Planning and Selection.

d. Feature V08: Physical Activity Spaces and Equipment.

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## V02 ERGONOMIC WORKSTATION DESIGN | P

**Intent :** Reduce the risk of physical strain on the body through ergonomic design at workstations that supports neutral body positions for seated and standing work and provides opportunities to alternate between seated and standing positions.

**Summary :** This WELL feature requires projects to provide ergonomic workstation furnishings to accommodate all users, that allow for customized workstation fit and provide user orientation to workstations covering ergonomic workstation design and adjustability features.

**Issue :** In 2016, musculoskeletal disorders (MSDs) ranked among the top drivers of global disability. [1,2] MSDs are one of the most commonly reported causes of lost or restricted work time and also contribute to absenteeism and low productivity.[3,4] Risk factors in the workplace vary by the type of tasks being performed. In manual labor work environments, risk factors include heavy lifting, bending, reaching overhead and pushing or pulling heavy objects.[4] In office settings, risk factors are no less prevalent and include workstation design that forces the body into awkward positions along with other occupational factors that expose the body to prolonged or repetitive tasks.[5]

**Solutions :** An ideal ergonomic work environment is conducive to the necessary breadth of tasks assigned to that space, while encouraging movement through a variety of positions throughout the day. Effective ergonomic interventions to accommodate all users include both design (e.g., adjustable furniture) and programmatic (e.g., education) approaches.[6,7] Ergonomic design solutions facilitate customizability at workstations allowing users to better fit workstations to their needs. Preliminary studies have demonstrated an ROI for ergonomics interventions. One study found a return of $10 USD for every $1 USD invested.[8] A second study examining outcomes across 250 case studies found generally positive results, including a reduction in the number (49.5% across 37 studies) and cost (64.8% across 22 studies) of work-related MSDs, and also noted that the payback period was generally less than one year.[9]

### PART 1 SUPPORT VISUAL ERGONOMICS

*For Office Spaces:*

The project meets the following requirements:

a. Workstations where desktop computers are used provide support for user adjustability (monitor height, viewing angle, horizontal distance) through one of the following:
   1. Monitors with built-in height and angle adjustment.[10,11]
   2. Monitor stands or arms that allow height, angle and horizontal adjustment.[10,11]

b. Workstations where laptops are primarily used provide support for user adjustability through at least one of the following:
   1. An external keyboard, mouse and laptop stand such that the laptop screen can be positioned by the user (screen height, viewing angle, horizontal distance).[11]
   2. An external monitor that meets requirement a.[11]

### PART 2 PROVIDE HEIGHT-ADJUSTABLE WORK SURFACES

*For Office Spaces:*

At least 25% of all workstations can be adjusted by the user for both seated and standing work, through one of the following:

a. Manual or electric height-adjustable work surfaces that provide users with the ability to customize workstation height at both seated and standing positions.[10,11]

b. Supplemental solutions (e.g., stand) that allow all or part of the work surface, monitor and primary input devices (e.g., keyboard, mouse) to be raised or lowered to seated or standing heights.[10,11]

### PART 3 PROVIDE CHAIR ADJUSTABILITY

*For Office Spaces:*

All seating at workstations meets the following requirements:

a. The seat height is adjustable.[10,11]

b. One of the following:[10]
   i. The seat pan depth is adjustable.
   ii. The seat pan depth is 43 cm or less.

c. One of the following:[10,11]
   i. The backrest lumbar support is height adjustable.
   ii. The backrest angle is adjustable.
   iii. The armrest height and distance between armrests are adjustable.

### PART 4 PROVIDE SUPPORT AT STANDING WORKSTATIONS

*For All Spaces except Commercial Kitchen Spaces:*

*Option 1: Support for standing workers*

All workstations in which users are regularly required to stand for 50% or more of their working hours (e.g., assembly line station, hotel check-in counter, supermarket check-out counter) incorporate at least two of the following:

a. Anti-fatigue mats, impact reducing flooring or a similar strategy.[12]

b. Recessed toe space of at least 10 cm depth and height.13

c. A footrest or footrail.[12,14]

d. A sit-stand stool or high stool.[12,14]

OR

*Option 2: No standing workers*

The project meets the following requirement:

a. There are no workstations at which users are regularly required to stand for 50% or more of their working hours.

### PART 5 PROVIDE WORKSTATION ORIENTATION

*For All Spaces:*

The following requirement is met:

a. All eligible employees receive an orientation (e.g., in-person or virtual training, self-guided digital training) to workstations in the space covering, at minimum, the following:
    1. Adjustability features of all available workstation types (as applicable) and their benefits to users (e.g., customized fit for individual comfort).
    2. Instructions on how to make adjustments to achieve the intended benefits (e.g., customized fit for individual comfort).
    3. Orientation resources that can be used for future reference.

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## V03 CIRCULATION NETWORK | O (MAX : 3 PT )

**Intent :** Encourage stair use through aesthetic design, signage and visibility of staircases.

**Summary :** This WELL feature requires projects to design staircases for everyday use and leverage aesthetics, visibility/positioning, and prompts to encourage stair use.

**Issue :** Physical inactivity and sedentariness have emerged as a primary focus of public health in recent years, due to the host of negative health implications associated with both behaviors.[1–6] Strategies that promote stair use and general movement throughout buildings have emerged as promising interventions that encourage short bouts of health-enhancing physical activity throughout the day.[7,8]

**Solutions :** Evidence from systematic reviews (including international data from a variety of settings, such as airports, healthcare facilities, universities and offices) suggests that stairwell enhancements and signage increase stair use.[9–12] In particular, point-of-decision design prompts that include directional signage and motivational messaging have been shown to be effective at increasing stair use.[9–14] Evidence suggests that improving aesthetic and atmosphere with design, music and artwork, as well as tailoring motivational signage and prompts to the audience or population the space serves, may help increase intervention effectiveness.[9,15,16] Novel strategies, such as gamification, which leverages game elements to encourage desired behaviors, have been introduced as fun and innovative ways to encourage healthy behaviors, such as stairclimbing in both public and private settings.[17] At an infrastructural level, considering where stairs are placed, which is particularly relevant for new construction projects, can have a large impact on movement opportunities throughout the day. Evidence-based guidelines, such as the Active Design Guidelines suggest that stairs should be proximate to main entry points and located physically and visibly before elevator or escalator banks.[7]

### PART 1 DESIGN AESTHETIC STAIRCASES (MAX : 1 PT )

*For All Spaces:*

At least one staircase is open to regular occupants, services all occupiable floors of the project and is aesthetically designed through the inclusion of at least two independent strategies from the following list on each floor:

a. Music.[7]

b. Artwork.[7]

c. Designed to have light levels of at least 9fc when in use.[19]

d. Windows or skylights that provide access to daylight and/or nature views.[7,18]

e. Natural design elements (e.g., plants, water features, images of nature).[7]

f. Gamification.[17]

**Note :** Interiors projects may earn this feature either through stairs within the project boundary or through base building stairs outside of the project boundary. Stairs must service all floors of the project but do not need to be accessible from the ground floor of the building.

### PART 2 INTEGRATE POINT-OF-DECISION SIGNAGE (MAX : 1 PT )

*For All Spaces:*

At least one staircase is open to regular occupants, services all occupiable floors of the project and is supported by the following:

a. Motivational, point-of-decision signage is present at the following locations:
   1. Near the main building entrance or the reception desk.[7]
   2. At elevator or escalator banks on each floor.[7]
   3. At the base of stairs and stairwell re-entry points on each floor.[7]

b. If stairs are not visible from signage locations, wayfinding signage is used to guide occupants to the stairs.[7]

**Note :** Interiors projects may earn this feature either through stairs within the project boundary or through base building stairs outside of the project boundary. Stairs must service all floors of the project but do not need to be accessible from the ground floor of the building. If the project does not access the stairs from the ground floor, they are exempt from requirement a1, which requires signage on the ground floor.

### PART 3 PROMOTE VISIBLE STAIRS (MAX : 1 PT )

*For All Spaces:*

At least one staircase is open to regular occupants, services all occupiable floors of the project and meets the following requirement:

a. Is at least as prominent (visual or physical) as elevators/escalators relative to the main point of entry to the building or project space (e.g., entrance to the project on their floor).[7,18]

**Note :** Interiors projects may earn this feature either through stairs within the project boundary or through base building stairs outside of the project boundary. Stairs must service all floors of the project but do not need to be accessible from the ground floor of the building.

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## V04 FACILITIES FOR ACTIVE OCCUPANTS | O (MAX : 3 PT )

**Intent :** Foster active commuting through facilities that support cycling to the building and active occupants more broadly.

**Summary :** This WELL feature requires projects to provide no-cost bike storage along with showers, changing facilities and lockers, which support both active commuters and active occupants.

**Issue :** Cycle commuting is associated with lower cardiovascular disease incidence and mortality, cancer incidence and all-cause mortality.[1] From 2000-2016, federal funding in the U.S. for pedestrian and cyclist infrastructure increased from 0.1% to 2.2% of transportation spending, which translated to an increase in those walking and biking to work.[2] However, research suggests that many communities still lack sufficient funding and infrastructure to support active commuting intentions.[2] For example, among safety concerns and pervasive societal norms of the car culture, lack of bike parking and post-commute facilities are key reasons why people opt not to cycle to the workplace.[3–8]

**Solutions :** In a comprehensive review of the literature on this topic, studies reported that availability of amenities, such as bike parking and showers had a positive impact on cycling. [9] Projections from one study showed that compared to baseline (5.8%), outdoor parking would increase those cycling to work to 6.3%, indoor, secure parking to 6.6% and indoor parking plus showers to 7.1%.[10] In addition, lockers and changing/shower facilities, in particular, support activity goals and behaviors not only for cyclists but all occupants, such as those who might engage in physical activity or exercise before work. Such amenities signal to occupants that physical activity and, in particular, active commuting, is welcomed and encouraged. Thoughtful site planning and selection can also enhance opportunities for cycling. The presence of cyclist infrastructure, such as cyclist lanes and, in particular, infrastructure that promotes cyclist safety, is known to increase ridership.[11–13]

### PART 1 PROVIDE CYCLING INFRASTRUCTURE (MAX : 2 PT )

*For All Spaces except Dwelling Units & Retail Spaces:*

*1: Cycling network*

One of the following requirements is met:

a. The project's address achieves a minimum Bike Score® of 50.[14] To utilize this tool, projects must be located in a country that is "supported" by the tool.

b. The building is within a 200 m walk distance of an existing cycling network that connects riders to at least 10 use types that are within a 4.8 km cycling distance of the project boundary.[15] Uses and restrictions are defined in Appendix V1.[16]

c. The project demonstrates existing plans for a cycling network that meets requirements a or b.

*2: Bike parking*

The following requirements are met:

a. Bike parking is provided to occupants at no cost in the following quantities:

   1. Short-term bike parking (e.g., public bike rack) is located within a 30 m walk distance of a functional building entrance and can accommodate at least 2.5% of peak visitors (minimum of four spaces per building).[15]

   2. Long-term bike parking (e.g., bike room) is available within the project boundary and can accommodate at least 5% of regular occupants, excluding occupants under eight years old (minimum of four spaces per building).[15]

b. The project provides no cost access to basic bike maintenance tools (e.g., bike pump and patch kit) co-located with long-term bike parking or quarterly on-site bike maintenance services.

**Note :** Interiors projects may count base building amenities towards feature requirements.

*For Retail Spaces:*

*1: Cycling network*

One of the following requirements is met:

a. The project's address achieves a minimum Bike Score® of 50.[14] To utilize this tool, projects must be located in a country that is "supported" by the tool.

b. The building is within a 200 m walk distance of an existing cycling network that connects riders to at least 10 use types that are within a 4.8 km cycling distance of the project boundary.[15] Uses and restrictions are defined in Appendix V1.[16]

c. The project demonstrates existing plans for a cycling network that meets requirements a or b.

*2: Bike parking*

The following requirements are met:

a. Bike parking is provided to occupants at no cost in the following quantities:

   1. Short-term bike parking (e.g., public bike rack) is located within a 30 m walk distance of a functional building entrance and includes at least two short-term bike storage spaces per 465 m2 of floor area (minimum of two

spaces per building).17

2. Long-term bike parking (e.g., bike room) is available within the project boundary and can accommodate at least 5% of regular occupants (minimum of two spaces per building).17

b. The project provides no cost access to basic bike maintenance tools (e.g., bike pump and patch kit) co-located with long-term bike parking or quarterly on-site bike maintenance services.

**Note** : Interiors projects may count base building amenities towards feature requirements.

*For Dwelling Units:*

*1: Cycling network*

One of the following requirements is met:

a. The project's address achieves a minimum Bike Score® of 50.[14] To utilize this tool, projects must be located in a country that is "supported" by the tool.

b. The building is within a 200 m walk distance of an existing cycling network that connects riders to at least 10 use types that are within a 4.8 km cycling distance of the project boundary.[15] Uses and restrictions are defined in Appendix V1.[16]

c. The project demonstrates existing plans for a cycling network that meets requirements a or b.

*2: Bike parking*

The following requirements are met:

a. Bike parking is provided to occupants at no cost in the following quantities:

1. Short-term bike parking (e.g., public bike rack) is located within a 30 m walk distance of a functional building entrance and can accommodate at least 2.5% of peak visitors (minimum of four spaces per building).15

2. Long-term bike parking (e.g., bike room) is located within the project boundary and can accommodate at least 30% of regular occupants (minimum of one space per building).15

b. The project provides no cost access to basic bike maintenance tools (e.g., bike pump and patch kit) co-located with long-term bike parking or quarterly on-site bike maintenance services.

**Note** : Interiors projects may count base building amenities towards feature requirements.

## PART 2 PROVIDE SHOWERS, LOCKERS AND CHANGING FACILITIES (MAX : 1 PT )

*For All Spaces except Dwelling Units & Guest Rooms:*

The following requirements are met:

a. Showers with changing facilities are available to occupants at no cost in a quantity listed below within a 200 m walk distance of the project boundary:[15]

| Regular Occupants (age 12 or older) | Required Number of Showers |
|---|---|
| 0 - 100 | One |
| 101 - 999 | One plus one for every 150 occupants above 100<br>Calculation: 1 + (# of occupants - 100) / 150 |
| 1,000 – 4,999 | Eight plus one for every 500 occupants above 1,000<br>Calculation: 8 + (# of occupants - 1,000) / 500 |
| 5,000 + occupants | 16 plus one for every 1,000 occupants above 5,000<br>Calculation: 16 + (# of occupants - 5,000) / 1,000 |

b. At least five lockers are available for every shower provided by the project. Lockers are co-located (i.e., in the same area/room, not on separate floors) with shower facilities.

**Note** : Interiors projects may count base building amenities towards feature requirements.

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## V05 SITE PLANNING AND SELECTION | O (MAX : 4 PT )

**Intent :** Promote movement, physical activity and active living through site and nearby amenities that facilitate walkability and provide proximate access to public transportation.

**Summary :** This WELL feature requires projects to demonstrate that the area around the building is fostering walkability and that the building is located near public transportation.

**Issue :** Over time, nearly every aspect of our built environment has been designed to demand and invite less movement – giving preference to sedentary activities, such as driving. [1] In addition to interior active design, the context in which a WELL project is situated, including neighborhood and site-level factors, plays an integral role in physical activity opportunities and choices.[2–6]

**Solutions :** The impact of thoughtful site planning, design and selection reaches beyond positive impacts on physical activity and active living, improving nearly every aspect of community health and vitality from social well-being to economic development.[7–9] There is no single metric or recipe that defines a walkable community. Features of walkable neighborhoods vary throughout the literature but centralize around several core design themes: proximity, connectivity, density, safety and aesthetics.[10] Walkable communities consider the needs of diverse users and abilities and are designed to facilitate mobility across the lifespan. Communities can be evaluated at different scales, down to the street and building scale. Single buildings can actually have important contributions to the streetscape. Buildings that activate the first level by incorporating aesthetic design can make positive contributions to the pedestrian environment.[11,12]

### PART 1 SELECT SITES WITH PEDESTRIAN-FRIENDLY STREETS (MAX : 2 PT )

*For All Spaces:*

*1: Pedestrian-friendly streets*

At least one functional building entrance opens to a pedestrian network (i.e., streets where pedestrians travel featuring, at a minimum, sidewalks) and meets at least one of the following requirements:

a. The project's address achieves a minimum Walk Score® of 70. To utilize this tool, projects must be located in a country that is "supported" by the tool.
b. Opens onto a street with restricted vehicular traffic.[14]
c. Within a 400 m distance of the project boundary, 90% of the total street length has continuous sidewalks on both sides and two of the following are met:
    1. At least eight existing use types (as defined in Appendix V1) are present.[15,16]
    2. Speed limits of 40 kmh or less and street has buffer protections along sidewalks (e.g., curb extension, bioswales, bike lane, parked cars, benches, trees, planters).[17-19]
    3. Street segments intersect one another (excluding alleys) at least every 80-100 m.[16,17]

*2: Pedestrian-friendly environment*

Exterior building walls facing the pedestrian network incorporate one or more of the following on the first floor or first 5.5 vertical m (whichever is less):[17,20,21]

a. Windows or glazing that provide transparency into the space.
b. Overhangs such as canopies, awnings, eaves or shades.
c. Murals or other artistic installations.
d. Biophilic design elements (e.g., plants, water features, nature patterns, natural building materials).
e. Mixed building textures, colors and/or other design elements.

**Note** : Interiors projects may count base building amenities toward feature requirements.

### PART 2 SELECT SITES WITH ACCESS TO MASS TRANSIT (MAX : 2 PT )

*For All Spaces:*

The project meets at least one of the following requirements:[22]

a. The project's address achieves a minimum Transit Score® of 70. To utilize this tool, projects must be located in country that is "supported" by the tool.
b. Is located within a 200 m walk distance of existing bus network that provides at least 72 trips on each weekday and 30 trips on each weekend day.
c. Is located within a 400 m walk distance of existing bus rapid transit stops, light or heavy rail stations, commuter rail stations or ferry services that provide at least 72 trips on each weekday and 30 trips on each weekend day.

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## V06 PHYSICAL ACTIVITY OPPORTUNITIES | O (MAX : 2 PT )

**Intent :** Encourage physical activity and exercise through no-cost physical activity opportunities for occupants.

**Summary :** This WELL feature requires projects to provide no-cost physical activity opportunities led by a qualified physical activity professional.

**Issue :** Nearly a quarter of the global population fails to achieve physical activity guidelines and is considered physically inactive. [1] Key determinants of physical activity behavior include time, convenience, motivation, self-efficacy, weather conditions, travel and family obligations, fear of injury, lack of social support and environmental barriers such as availability of sidewalks, parks and bicycle lanes.[2,3] In a review conducted by the Centers for Disease Control and Prevention, two studies highlighted economic benefits of workplace programs, including reduced healthcare costs, decreased costs and days lost due to disability, reduced absenteeism and increased productivity.[4]

**Solutions :**
The workplace is considered an effective platform to reach a broad segment of the adult population.[5] Therefore, workplace wellness programs and offerings are considered great steps toward decreasing barriers to physical activity engagement among employees.[6] The Community Preventive Services Task Force recommends worksite programs that make physical activity more readily available (e.g., providing health club memberships, changing insurance benefits, providing opportunities to be physically active) as a strategy to improve physical activity engagement.[7]

Similar to the workplace, schools represent a ubiquitous platform to reach adolescents and youth.[5] The Community Preventive Services Task Force recommends classroom-based teaching strategies and physical education curricula that incorporate activity as promising strategies to increase physical activity among adolescents.[8] When considering physical activity education and programming, it is vital to consider the needs of the population that the project serves. Events and education should be relevant to the community (i.e., ability and age-appropriate). Projects should also seek to solicit on-going feedback from their population and make an effort to consider feedback in revisions to programmatic offerings.

### PART 1 OFFER PHYSICAL ACTIVITY OPPORTUNITIES (MAX : 2 PT )

*For All Spaces:*

No cost physical activity opportunities are available to regular occupants and meet the following requirements:

a. Programming is appropriate for the project population (e.g., age, ability, culture).
b. Programming is offered by a qualified physical activity professional either in-person within a 200 m walk distance of the project boundary or virtually through live programming.
c. As applicable, physical activity opportunities are not withheld as a form of punishment for early childhood education, primary or secondary school students.[9]
d. Programming is offered at the following frequencies, as applicable:

| Tier | Early Childhood Education, Primary and Secondary School Students (as applicable) | All other regular occupants | Points |
|---|---|---|---|
| 1 | At least one 60-minute event per week | At least one 30-minute event per week | 1 |
| 2 | > 60 minutes per school day[10] | > 150 minutes per week[10] | 2 |

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## V07 ACTIVE FURNISHINGS | O (MAX : 2 PT )

**Intent :** Encourage movement, postural breaks and switching and discourage prolonged sitting or standing at stationary workstations through active workstations.

**Summary :** This WELL feature requires projects to provide ample active workstations, such as a sit-stand or treadmill desk.

**Issue :** Sedentary behavior has been linked to numerous negative health outcomes, including obesity, type 2 diabetes, cardiovascular and metabolic risks and premature mortality. [1–5] Sedentary behavior also poses health risks, despite activity levels, and may even negate the positive health effects associated with physical activity.[1,6–8]

**Solutions :** Active workstations have grown in popularity in commercial office environments in recent years. Active workstations are effective at decreasing time spent sitting, thereby increasing energy expenditure.[9–13] Studies do not suggest there is an impact on productivity for sit-stand or treadmill desks with more mixed findings for bicycle desks. [11,12,14–18] Evidence further suggests that offering active workstations along with education, prompts and/or behavior change counseling may support sustained behavior change and further reduce sitting time.[19–21]

### PART 1 PROVIDE ACTIVE WORKSTATIONS (MAX : 2 PT )

*For Office Spaces:*

Active workstations meet the following requirements:

a. Enable users to sit/stand or obtain light physical activity during use through one of the following:

i. Manual or electric height adjustments for work surface.

ii. Treadmill component.

iii. Stationary bicycle component.

iv. Step machine component.

b. Are provided in one of the following quantities:

| Tier | Active Workstation Quantity | Points |
|---|---|---|
| 1 | At least 50% of workstations | 1 |
| 2 | At least 90% of workstations | 2 |

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## V08 PHYSICAL ACTIVITY SPACES AND EQUIPMENT | O (MAX : 2 PT )

**Intent :** Promote physical activity and exercise by providing access to physical activity spaces and equipment at no cost.

**Summary :** This WELL feature requires projects to provide access to a physical activity space at no cost through an on-site fitness facility, nearby facility or nearby outdoor spaces, such as a park.

**Issue :** International physical activity guidelines recommend both cardiovascular and muscle strengthening activities for the general population.[1] Despite widely disseminated guidelines, nearly a quarter of the general population fails to achieve recommended physical activity levels.[2] Key determinants of physical activity behavior include time, convenience, motivation, self-efficacy, weather conditions, travel and family obligations, fear of injury and lack of social support.[3,4] At a community scale, additional environmental barriers exist, such as availability of sidewalks, parks and bicycle lanes.[3,4]

**Solutions :** In a systematic review conducted by the U.S. Centers for Disease Control and Prevention, creating enhanced places for physical activity increased engagement and biomarkers for physical fitness, including aerobic capacity and energy expenditure, with a few studies documenting some decrease in body weight and body fat.[5] Creating space within a building for physical activity is important, along with larger efforts to design communities to encourage movement. Factors such as proximity and quality of parks are important predictors of physical activity, although more consistent research is needed.[6,7]

### PART 1 PROVIDE INDOOR ACTIVITY SPACES (MAX : 1 PT )

*For All Spaces:*

*Option 1: On-site physical activity spaces*

A dedicated fitness facility is available within the project boundary at no cost to regular occupants and is sized according to one of the following requirements:

a. The space includes at least two types of exercise or sporting equipment (e.g., free weights, treadmill, yoga mat, basketball) in quantities that allow use by at least 5% of regular occupants at any time.[8]

b. The space includes at least two types of exercise or sporting equipment (e.g., free weights, treadmill, yoga mat, basketball). The minimum size is $25 m^2$ plus $0.1 m^2$ per regular occupant or $930 m^2$, whichever is smaller.[9]

OR

*Option 2: Off-site physical activity spaces*

The project meets the following requirement:

a. Regular occupants have access to a fitness facility at no cost within a 200 m walking distance of the project boundary.

### PART 2 PROVIDE OUTDOOR PHYSICAL ACTIVITY SPACE (MAX : 1 PT )

*For All Spaces:*

At least one of the following outdoor physical activity spaces is within a 400 m walk distance of the project boundary and available at no cost to regular occupants:

a. Green space (e.g., park, walking/biking trail).

b. Blue space (e.g., swimming area).

c. Recreational field or court.

d. Fitness zone that includes all-weather fitness equipment.

e. For projects with child occupants, play space geared toward children (e.g., playground).

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## V09 PHYSICAL ACTIVITY PROMOTION | O (MAX : 1 PT )

**Intent :** Encourage physical activity and exercise, by designing, implementing and monitoring physical activity incentive programs.

**Summary :** This WELL feature requires projects to provide physical activity incentives or promotion programs and monitor uptake of offerings.

**Issue :** Physical inactivity has emerged as a primary focus of public health, due to a rise in premature mortality and chronic diseases attributed to inactive lifestyles, including type 2 diabetes, cardiovascular disease, depression, stroke and some forms of cancer.[1,2] Physical activity is intimately tied to prevention of these chronic conditions and overall health across the lifespan.[3] In adolescent populations, physical activity is especially important for health, well-being and development.[3]

**Solutions :**
Health promotion programs in the workplace typically take a comprehensive approach to behavior change, including environmental design and behavioral strategies. [3–5] There isn't a templatized program that can be applied and deliver results in every workplace.[3] It is, therefore, critical that projects take an integrative approach and design a program that is tailored to the needs of the population they serve. Physical activity incentives are one of many strategies that can be used. In a systematic review examining different incentives, conditional incentives – particularly those that rewarded positive physical activity behavior – were effective at improving physical activity levels as compared to unconditional incentives (e.g., subsidized memberships that do not require participation).[6]

Just as workplaces are a ubiquitous platform to reach working populations, programs targeting schools are a key avenue to reach student populations.[3] In schools, some of the most common and well-recognized strategies to provide physical activity opportunities are through physical education curricula, recess, afterschool programs and sport, and activation of classrooms to incorporate more movement.[3,7–9] It's also recommended that schools incorporate programs that address sedentary behavior, such as screen time, as part of a comprehensive approach to health promotion.[10]

### PART 1 OFFER PHYSICAL ACTIVITY INCENTIVES (MAX : 1 PT )

*For All Spaces:*

*1: Incentives for eligible employees*

The project or organization offers at least two of the following physical activity promotion programs to eligible employees:

a. Rewards for physical activity engagement (e.g., prizes, financial rewards).
b. A subsidy towards physical activity costs incurred by employees (e.g., membership fees or group fitness classes), including those incurred during business travel.
c. Reductions in health care premiums based on physical activity engagement.
d. Flexible work hours to accommodate physical activity.
e. Paid time off for physical activity with a minimum of four days per calendar year. Days must be used towards physical activity engagement or recovery and may not be deducted from regular paid time off or other employer-provided time off from work (e.g., sick leave, standard paid holidays).

*2: Employee utilization of incentive programs*

One of the following requirements is met:

a. The project or organization monitors utilization of incentive programs and demonstrates an annual utilization rate of 50% (i.e., at least 50% of eligible employees have utilized at least one incentive over the past year). Projects may report combined utilization rates across multiple incentives, as appropriate.
b. The project or organization demonstrates an annual improvement in utilization of at least 10 percentage points. Projects may report combined utilization rates across multiple incentives, as appropriate.

*3: Physical activity for students*

Early childhood education, primary and secondary schools develop and implement the following programs for students:

a. A program that aims to reduce daily time spent in the following sedentary behaviors:
   1. TV viewing.
   2. Recreational computer or smartphone use.
   3. Gaming.
b. A program that aims to promote daily physical activity through at least one of the following:
   1. Teaching strategies that incorporate movement and activity into the lesson.
   2. Physical education.
   3. Recess or similar physical activity breaks.

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## V10 SELF-MONITORING | O (MAX : 1 PT )

**Intent :** Promote self-awareness of health behaviors and health metrics through wearable technology.

**Summary :** This WELL feature requires projects to provide or subsidize wearables that can monitor physical activity and health behaviors over time.

**Issue :** Much of what we understand about physical activity's relationship to health and well-being is derived from studies that utilize self-reported measures of physical activity. [1] Evidence suggests that self-reported measures tend to overestimate actual physical activity behaviors.[2]

**Solutions :** Objective, accelerometer-based tools that track physical activity have proliferated the marketplace.[3–5] Estimates from 2016 indicate 33% of people across 16 countries (including 20,000 individuals) used a wearable or app to track physical activity and health parameters.[3,6,7] In addition, an estimated 44.5% of employers leverage them in strategic planning of wellness programs.[8] Evidence from a systematic review conducted by the U.S. Centers for Disease Control and Prevention shows that, particularly when paired with coaching and counseling, technology tools can have a positive impact on recording reliable data and improving health outcomes.[9] In a comprehensive review led by the U.S. Physical Activity Guidelines Committee, researchers found evidence that wearable activity monitors, including simple step counters, when paired with goal-setting, were effective at increasing physical activity.[10] In a meta-analysis of several randomized controlled trials, use of a step counter paired with a step goal significantly reduced sedentary time among adults.[11] Projects should consider privacy and data security among users when they are vetting technologies to recommend to their occupants, and when projects may have access to users' wearable data.

### PART 1 PROVIDE SELF-MONITORING TOOLS (MAX : 1 PT )

*For All Spaces:*

The project or organization provides devices (e.g., wearable fitness tracker) to all eligible employees that meet the following requirements:

a. Available at no cost or subsidized by at least 50%.

b. Allow users to monitor their own metrics over time (i.e., provides a dashboard where individual metrics are aggregated).

c. Measure at least two physical activity metrics (e.g., steps, floors climbed, activity minutes).

d. Measure at least one additional health behavior (e.g., mindfulness practice, sleep).

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## V11 ERGONOMICS PROGRAMMING | O (MAX : 3 PT )

**Intent** : Enhance well-being and comfort through comprehensive ergonomics programming.

**Summary** : This WELL feature requires projects to work with a certified ergonomist to implement comprehensive ergonomics programming, commit to on-going improvements to ergonomic design and provide ergonomic support for remote workers.

**Issue** : In 2016, musculoskeletal disorders (MSDs) ranked among the top drivers of global disability. [1,2] MSDs are one of the most commonly reported causes of lost or restricted work time and also contribute to absenteeism and low productivity among employees.[3,4] These losses amount to significant impacts for the economy at large and a business's bottom line. [5] Ergonomic issues affect many sectors from schools to industrial settings and commercial offices, each with unique, contextual risks and considerations.[6]

**Solutions** : Ergonomic interventions aim to accommodate all individuals and increasingly adopt a holistic approach that tackles the design of the physical environment (e.g., adjustable furniture), the work itself (e.g., process, practices) and behavior (e.g., education, training).[7,8] Ergonomic interventions have been shown to have a positive impact and a substantial return on investment (ROI). One study found that after implementing a macro-ergonomics intervention, claims over a 5-year period were reduced by 45% (200 fewer total claims) and researchers determined an ROI of 10:1 for the program. In this study, ROI calculations considered the compensation costs per claim, the number of preventive ergonomic evaluations performed and the annual cost of the program.[9] Another study examining outcomes across 250 case studies, across a variety of sectors from healthcare (36) to offices (40), manufacturing facilities (87) and other industries, found generally positive results and noted that the payback period was generally less than one year.[10]

### PART 1 IMPLEMENT AN ERGONOMICS PROGRAM (MAX : 1 PT )

*For All Spaces:*

*1: Professional ergonomics support*

The project or organization meets at least one of the following requirements:

a. Engages with a qualified professional ergonomist (which may be a consultant, contractor or other third-party).
b. Has at least one qualified professional ergonomist on staff whose responsibilities include ergonomic programming, as defined in their job description and/or performance expectations.

*2: Ergonomics programming*

The project or organization has an ergonomics program that includes the following:

a. Annual consultation with key stakeholders (e.g., human resources, workplace wellness, occupational safety, leadership, employees) who are involved in the design, implementation and evaluation of the ergonomics program.
b. A task analysis performed by a qualified professional ergonomist to identify the job-related tasks that are performed by occupants in the space.
c. Accommodations for eligible employees to receive individual ergonomic assessments in the form of a self-assessment (e.g., reputable, third-party app), in-person assessment (e.g., at the workplace or home) or a virtual assessment (e.g., live video conference). Assessments are offered to employees at least annually and, as applicable, during or after the following events:
   1. Employee on-boarding.
   2. Substantial equipment changes (e.g., purchase of a new chair) or workstation redesign.
   3. Change in health status (e.g., injury, presentation of symptoms of musculoskeletal issues, visual strain).
   4. Change in work environment (e.g., transition to or from full-time remote work).
d. Ergonomic trainings (e.g., workshop, seminar, classes) delivered by a qualified professional ergonomist to employees at least annually.

### PART 2 COMMIT TO ERGONOMIC IMPROVEMENTS (MAX : 1 PT )

**Note** :

Projects may only achieve this part if Part 1 is also achieved.

*For All Spaces:*

*Option 1: Informed Ergonomic Design*

The following requirement is met:

a. The project or organization describes how Part 1 informed design-decisions within Feature V02: Ergonomics Workstation Design and, as applicable, Feature V07: Active Furnishings.

OR

*Option 2: Individual Ergonomic Needs*

The following requirement is met:

a. The project or organization demonstrates a commitment to addressing the individual ergonomic needs of employees identified through individual ergonomics assessments.
b. The timeline for delivery of solutions is communicated to employees.

### PART 3 B SUPPORT REMOTE WORK ERGONOMICS (MAX : 1 PT )

**Note** :

Projects may only achieve this part if Part 1 is also achieved.

*For All Spaces:*

For organizations with employees who regularly work remotely, the organization tailors the ergonomic program addressed in Part 1 in the following ways:

a. For individual assessments, accommodations are made to include remote workers (e.g., virtual assessments).

b. The organization provides ergonomic equipment to support the individual needs of remote workers through direct-purchases, reimbursement or subsidies.

**Note :** This feature is a beta strategy and has an additional documentation requirement (beta feature feedback form). The feedback form supports IWBI in developing new features that are effective and applicable to projects around the world.

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## APPENDIX V1:

Diverse use types:[1]

| Category | Use type |
| --- | --- |
| Food retail | Supermarket |
| | Grocery with a produce section |
| Community-serving retail | Convenience store |
| | Farmer's market |
| | Hardware store |
| | Pharmacy |
| | Other retail |
| Services | Bank |
| | Family entertainment venue (e.g., theater, sports) |
| | Gym, health club, exercise studio |
| | Hair care |
| | Laundry, dry cleaner |
| | Restaurant, café or diner (excluding those with only drive-thru service) |
| Civic and community facilities | Adult or senior care (licensed) |
| | Child care (licensed) |
| | Community or recreation center |
| | Cultural arts facility (museum, performing arts) |
| | Education facility (e.g., K–12 school, university, adult education center, vocational school, community college) |
| | Government office that serves public on-site |
| | Medical clinic or office that treats patients |
| | Place of worship |
| | Police or fire station |
| | Post office |
| | Public library |
| | Public park |
| | Social services center |
| Community anchor uses | Commercial office (100 or more full-time employees) |
| | Housing (100 or more dwelling units) |

The following restrictions apply to Appendix V1:

1. A use may be counted as only one use-type (e.g., a single retail space may be counted only once, even if it sells products in several use categories).
2. No more than two uses in each use type may be counted (e.g., if five restaurant spaces are within the required distance, only two may be counted).

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